

D692

Early Mathematics Methods and Interventions

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A pre-K teacher with a colorful math rug asks a child to count 7 teddy bears. The child touches each bear once and says “1, 2, 3, 4, 5, 6, 7.” When asked “How many bears are there?” the child starts counting again. The child’s older brother is in second grade. What does this show?

- A. The child needs to learn numbers to 20 first.
- B. The child lacks one-to-one correspondence.
- C. The child has one-to-one correspondence but not yet cardinality.
- D. The child fully understands counting.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which behavior shows that a child understands cardinality?

- A. The child recounts every time to be sure.
- B. After counting 5 blocks, the child says “5” when asked how many, without recounting.
- C. The child touches each object once while counting.
- D. The child recites numbers 1 to 30 in order.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A kindergarten teacher who uses a puppet named Max flashes a dot card for two seconds showing 3 dots on top and 3 dots below. A child quickly says, “Six — three and three.” The class has snack right after math. What skill is the child showing?

- A. Rote counting done quickly.
- B. Perceptual subitizing.
- C. Counting on.
- D. Conceptual subitizing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A first grader who is very good at building with blocks can solve $14 - 6$ correctly with base-ten blocks but cannot solve $14 - 6$ when it is written on a worksheet. His teacher recently moved from teaching third grade. Under the concrete-representational-abstract approach, what is the best next step?

- A. Have him draw quick pictures of tens and ones to solve, linking each drawing to the written equation.
- B. Keep using only blocks for the rest of the year.
- C. Give more worksheets until he gets it.
- D. Take the blocks away permanently.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

In a first-grade class with new magnetic ten frames, a student solves $8 + 5$. The class earned a sticker chart reward that morning. Which reasoning shows the make-ten strategy?

- A. "9, 10, 11, 12, 13."
- B. " $8 + 2$ is 10, and 3 more is 13."
- C. " $5 + 8$ is the same as $8 + 5$."
- D. " $8 + 2$ is 10, and 5 more is 15."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A student using make-ten says, " $7 + 6 = 7 + 3 + 3$." What should the next step be?

- A. $10 + 3 = 13$
- B. $7 + 3 = 10$, so the answer is 10.
- C. $7 \times 6 = 42$
- D. $10 + 6 = 16$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A second-grade teacher who coaches soccer gives three story problems about marbles. Most students solve two easily but struggle with: “Lin had some marbles. Her friend gave her 4 more. Now she has 11. How many did she have at first?” The marbles in the problems are glass. Why is this problem harder?

- A. It uses larger numbers than the others.
- B. It is a result-unknown problem.
- C. It is a start-unknown problem, so students cannot simply act it out from the beginning.
- D. Students are unfamiliar with glass marbles.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which story problem has the start unknown?

- A. “There were 8 birds. Some flew away. Now there are 5. How many flew away?”
- B. “Some birds were in a tree. 3 flew away. Now there are 5. How many were there at first?”
- C. “5 red birds and 3 blue birds are in a tree. How many birds?”
- D. “There were 8 birds. 3 flew away. How many are left?”

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

On a worksheet printed in green ink, a first grader fills in $8 + 4 = \underline{\quad} + 5$ with 12. The student is one of the fastest in the class at timed addition facts. What misconception does this show, and what is the correct answer?

- A. The student added wrong; the answer is 17.
- B. The student is too fast; slowing down will fix it.
- C. The student reads $=$ as “the answer comes next”; the correct answer is 7.
- D. No misconception; 12 is correct.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Which equation is true?

- A. $7 = 3 + 4$
- B. $3 + 4 = 7 + 2$
- C. $4 + 3 = 34$
- D. $5 + 2 = 7 + 1 = 8$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A second-grade team at a school with a new math coach wants to build addition fact fluency. One teacher, who has a stopwatch app she likes, proposes daily timed tests as the main approach. What is the best recommendation?

- A.** Skip fluency, since calculators are always available.
- B.** Have students memorize facts with no strategies.
- C.** Adopt daily timed tests as the main approach.
- D.** Teach and practice strategies such as doubles and make-ten, with games and low-stress practice, and assess accuracy, efficiency, and flexibility.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A first grader in a class with 20 students solves $9 + 4$ by counting out 9 cubes, then 4 cubes, then counting all 13 from 1. She gets correct answers but is very slow. Her favorite color is purple. Which intervention fits best?

- A.** Give her more of the same worksheets.
- B.** Use purple cubes so she enjoys counting all.
- C.** Move her to multiplication to challenge her.
- D.** Explicitly teach counting on from the larger number with a number path, then practice.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A pre-K teacher ~~with a colorful math rug~~ asks a child to count 7 teddy bears. The child touches each bear once and says “1, 2, 3, 4, 5, 6, 7.” When asked “How many bears are there?” the child starts counting again. ~~The child’s older brother is in second grade.~~ What does this show?

Target word: which counting principle is missing **Decides:** Correct counting, but no sense that the last number is the total.

A	Adjacent category	Rote counting further is a different skill.
B	Confusion set	The child touched each bear exactly once.
C	Correct	Touches each bear once, but does not link the last number to “how many.”
D	Dropped element	Drops the recount, which is the key evidence.

2. Answer B Concept 1 • transfer

Which behavior shows that a child understands cardinality?

Target word: evidence of cardinality — reverse of spotting its absence **Decides:** Answers “how many” with the last number counted.

A	Inverted operation	Recounting signals cardinality is missing.
B	Correct	The last count word names the total.
C	Confusion set	One-to-one correspondence, not cardinality.
D	Adjacent category	Rote counting (stable order) only.

3. Answer D Concept 2

A kindergarten teacher ~~who uses a puppet named Max~~ flashes a dot card for two seconds showing 3 dots on top and 3 dots below. A child quickly says, “Six — three and three.” ~~The class has snack right after math.~~ What skill is the child showing?

Target word: name the skill **Decides:** Sees parts and combines them without counting.

A	Inverted operation	Two seconds was too brief to count each dot.
B	Adjacent category	Perceptual is instantly seeing a small set as a whole, without combining parts.
C	Confusion set	No counting sequence was used.
D	Correct	Recognizes two small groups and combines them to name the total.

4. Answer A Concept 3

A first grader ~~who is very good at building with blocks~~ can solve $14 - 6$ correctly with base-ten blocks but cannot solve $14 - 6$ when it is written on a worksheet. ~~His teacher recently moved from teaching third grade.~~ Under the concrete-representational-abstract approach, what is the best next step?

Target word: next CRA step **Decides:** Concrete is secure; bridge to drawings before symbols.

A	Correct	The representational stage bridges objects and symbols.
B	Inverted operation	Never moves toward the abstract stage.
C	Dropped element	Drops the representational bridge.
D	Absolute language	Manipulatives stay available while links are built.

5. Answer B Concept 4

In a first-grade class ~~with new magnetic ten frames~~, a student solves $8 + 5$. ~~The class earned a sticker chart reward that morning.~~ Which reasoning shows the make-ten strategy?

Target word: make-ten reasoning **Decides:** Take 2 from 5 to fill 8 to 10, then add 3.

A	Adjacent category	Counting on — a valid strategy, but not make-ten.
B	Correct	Decomposes 5 into $2 + 3$ to complete a ten.
C	Confusion set	The commutative property; it does not find the sum.
D	Dropped element	Drops the step of taking 2 from the 5.

6. Answer A Concept 4 • transfer

A student using make-ten says, " $7 + 6 = 7 + 3 + 3$." What should the next step be?

Target word: continue the strategy **Decides:** $7 + 3$ makes 10, then add the remaining 3.

A	Correct	Completes the ten, then adds the leftover.
B	Inverted operation	Stops at ten, dropping the remaining 3.
C	Confusion set	Switches to multiplication.
D	Dropped element	Forgets that 3 was already taken from the 6.

7. Answer C Concept 5

A second-grade teacher ~~who coaches soccer~~ gives three story problems about marbles. Most students solve two easily but struggle with: "Lin had some marbles. Her friend gave her 4 more. Now she has 11. How many did she have at first?" ~~The marbles in the problems are glass.~~ Why is this problem harder?

Target word: why this problem is harder **Decides:** The start amount is unknown.

A	Wrong stem number	4 and 11 are small; the structure is the difficulty.
B	Inverted operation	The result (11) is given.
C	Correct	Start-unknown join problems are the hardest structure.
D	Camouflage	The material is irrelevant.

8. Answer B Concept 5 • transfer

Which story problem has the start unknown?

Target word: identify structure **Decides:** The beginning amount is missing.

A	Confusion set	Change unknown.
B	Correct	The starting amount is the unknown.
C	Adjacent category	Part-part-whole, whole unknown.
D	Inverted operation	Result unknown.

9. Answer C Concept 6

On a worksheet printed in green ink, a first grader fills in $8 + 4 = \underline{\quad} + 5$ with 12. ~~The student is one of the fastest in the class at timed addition facts.~~ What misconception does this show, and what is the correct answer?

Target word: equal-sign misconception **Decides:** Both sides must equal 12, so the blank is 7.

A	Inverted operation	Adds all numbers — another form of the same misconception.
B	Camouflage	Speed is not the issue; the meaning of = is.
C	Correct	$8 + 4 = 12$, and $7 + 5 = 12$.
D	Dropped element	Drops the + 5 on the right side.

10. Answer A Concept 6 · transfer

Which equation is true?

Target word: recognize a true equation **Decides:** Both sides have equal value.

A	Correct	Both sides equal 7; the answer can be on the left.
B	Dropped element	$7 \neq 9$; the + 2 was ignored.
C	Inverted operation	Joins the digits instead of adding.
D	Confusion set	A run-on string; $5 + 2$ does not equal 8.

11. Answer D Concept 7

A second-grade team ~~at a school with a new math coach~~ wants to build addition fact fluency. One teacher, ~~who has a stopwatch app she likes~~, proposes daily timed tests as the main approach. What is the best recommendation?

Target word: how to build fluency **Decides:** Fluency grows from strategies and understanding.

A	Absolute language	Fluency supports later math.
B	Dropped element	Drops understanding and flexibility.
C	Camouflage	Uses the stopwatch; speed drills do not build strategies and can raise anxiety.
D	Correct	Builds all parts of fluency from understanding.

12. Answer D Concept 8

A first grader ~~in a class with 20 students~~ solves $9 + 4$ by counting out 9 cubes, then 4 cubes, then counting all 13 from 1. She gets correct answers but is very slow. ~~Her favorite color is purple.~~ Which intervention fits best?

Target word: targeted intervention **Decides:** She counts all; the next strategy is counting on.

A	Dropped element	Drops diagnosis and new strategy.
B	Camouflage	Color does not change the inefficient strategy.
C	Adjacent category	Skips the foundation she needs.
D	Correct	Targets her specific gap with the next strategy.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Cardinality	Cardinality is understanding that the last number said when counting tells how many are in the set.	Assuming a child who counts correctly understands how many.	A child who recounts when asked “How many?”
2	Subitizing	Subitizing is seeing how many without counting — instantly for small sets (perceptual) or by combining parts (conceptual, “3 and 3 is 6”).	Calling fast counting subitizing.	Dot cards flashed briefly, ten frames, dice patterns.
3	Concrete-representational-abstract	Build understanding with objects first, then drawings or diagrams, then numbers and symbols, linking each stage to the next.	Starting with symbols, or using manipulatives with no bridge to symbols.	A student who can do it with blocks but not on paper, or the reverse.
4	Make-ten strategy	Decompose one addend to complete a ten, then add the rest: $8 + 5 = 8 + 2 + 3 = 13$.	Adding to make ten without taking that amount from the other addend.	A sum that crosses ten, with a ten frame nearby.
5	Problem structures	Join and separate problems can have the result, change, or start unknown; start-unknown problems are the hardest because the action cannot simply be acted out forward.	Treating all addition and subtraction story problems as equally hard.	A story problem where the missing number is the beginning amount.
6	The equal sign is relational	The equal sign means both sides have the same value, not “write the answer next.”	Filling $8 + 4 = _ + 5$ with 12.	An equation with operations on both sides or the answer on the left.
7	What fact fluency means	Fluency is accurate, efficient, and flexible use of strategies, built from understanding; timed tests alone do not build it.	Equating fluency with speed on timed drills.	A plan to “build fluency” with daily timed tests.
8	Diagnose before intervening	Effective early intervention identifies the specific strategy gap (e.g., counting all vs. counting on) and teaches the next strategy explicitly with visual models and practice.	Giving more of the same worksheets or just re-teaching louder.	A student’s repeated error pattern or slow, inefficient strategy.

Course scope (public description): Early Mathematics Methods and Interventions is a comprehensive course designed for educators focused on advancing mathematics education in the early elementary grades. This course emphasizes innovative and research-based teaching methods for developing mathematical understanding in young children, particularly in the crucial developmental stages in PK through the early elementary grades. Participants will explore a variety of instructional strategies to support conceptual understanding and procedural fluency in areas such as number sense, basic operations, and early problem-solving skills. A significant aspect of the course involves identifying and addressing learning needs through targeted interventions, personalized instruction, and the use of manipulatives and digital tools. The curriculum also highlights the importance of creating an engaging and inclusive learning environment that encourages exploration and curiosity in mathematics. By integrating assessment techniques and thoughtful learning experiences, this course aims to

provide educators with the skills to effectively nurture early mathematical abilities and lay a strong foundation for future academic success in mathematics.